

AC9M6P01 • Year 6 Mathematics

Probability on a Scale from 0 to 1

Compare events using fractions, decimals and percentages

We are learning to...

Students place impossible, unlikely, even-chance, likely and certain events on a numerical scale and convert among equivalent probability forms.

Represent

Reason

Calculate

Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

recognise that probabilities lie on numerical scales from 0 to 1 and use fractions, decimals and percentages to compare the likelihood of events

Locate probability values on one scale



Probability 0 means impossible under the model and 1 means certain. Values between express relative likelihood, not a guarantee of short-run frequency.

Connect fractions, decimals and percentages

fraction	decimal	percentage
1/4	0.25	25%
1/2	0.5	50%
3/5	0.6	60%
4/5	0.8	80%
1	1.0	100%

Compare probabilities in a common representation and ensure the event and chance model are clearly defined.

Build and annotate the model

Represent probability on a scale from 0 to 1 and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Probability treated as a count:** It is a relative value from 0 to 1.
- **80% written as 80:** Write 0.8 on the probability scale.
- **Likely confused with certain:** A likely event can fail.
- **Context assumptions omitted:** State the chance model.

Quick check

- Place 0.35 on scale.
- Convert $\frac{3}{5}$.
- Compare 45% and 0.5.
- Name impossible/certain.
- Explain likelihood not guarantee.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.